

Towards precision sleep medicine: Self-attention GAN as an innovative data augmentation technique for developing personalized automatic sleep scoring classification

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ABSTRACT

It is very important to have good quality sleep, which can affect aspects such as memory consolidation, emotional regulation, learning, physical development, and quality of life. Diagnosing human sleep quality and problems quickly and accurately is an important issue for human well-being. Therefore, many automatic sleep scoring methods have been proposed. However, the methods have been developed using sleep data from different individuals or groups. The accuracies of these proposed methods might decrease, due to existing individual differences. In this study, the self-attention generative adversarial network (SAGAN) was applied as an advanced data augmentation technique to propose an improved personalized automatic sleep scoring classification. First, the spectrograms were converted from electroencephalography (EEG). Then, SAGAN was used to generate synthesized spectrograms for each subject. Finally, the real and synthesized spectrograms of each subject were utilized to train a personalized classifier. The averaged accuracy and standard deviation of the proposed method are 95.74% and 3.78%, respectively. Compared to the classifier trained with all subjects' training data, the average accuracy increased by 8.08%. The results proved that the generated spectrograms significantly improved the performance of the personalized automatic sleep scoring classification. The contributions of the proposed method were that made the medical staff and subjects save massive medical resources and time for manual recording and scoring.

1. Introduction

To diagnose and treat sleep-related disorders, sleep stage classification with overnight polysomnography (PSG) recordings plays an essential role [1]. The PSG recordings consist of the electroencephalogram (EEG), electromyogram (EMG), electrooculogram (EOG), and electrocardiogram (ECG), etc. [2]. Clinically, the PSG recordings are divided into sequential 30-s (30s) epochs and then each epoch is labeled as one of the sleep stages by clinicians manually following the guidelines of the Rechtschaffen and Kales (R&K) [3] or the American Academy of Sleep Medicine (AASM) [4]. Regarding the AASM manual, the sleep stages can be defined as Wake, Rapid Eye Movement (REM), Non-REM1 (N1), Non-REM2 (N2) and Non-REM3 (N3), respectively. The characteristics of different sleep stages are shown in Fig. 1.

However, it is cumbersome, time-consuming and prone to be subjective errors for the manual sleep scoring [5]. The patient need to wait at least two months for PSG recording at the hospital or sleep center. Additionally, the structure of whole-night sleep is greatly related to the subject's physiological and psychological condition and data acquisition environment. From the clinical staff's point, the patient's sleep stage usually changes frequently such as the patient with apnea may experience abnormal breathing during sleep, which would often awaken the brain from deeper sleep that made the agreement between the experts is often not high as a result [6]. Moreover, people who have insomnia show a higher proportion of the stage N1 that is the most challenging to be recognized. Generally, an all-night PSG recording (approximately 6–8 h) usually take at least 1 h for manual scoring.

In the past twenty years, various automatic sleep stage scoring

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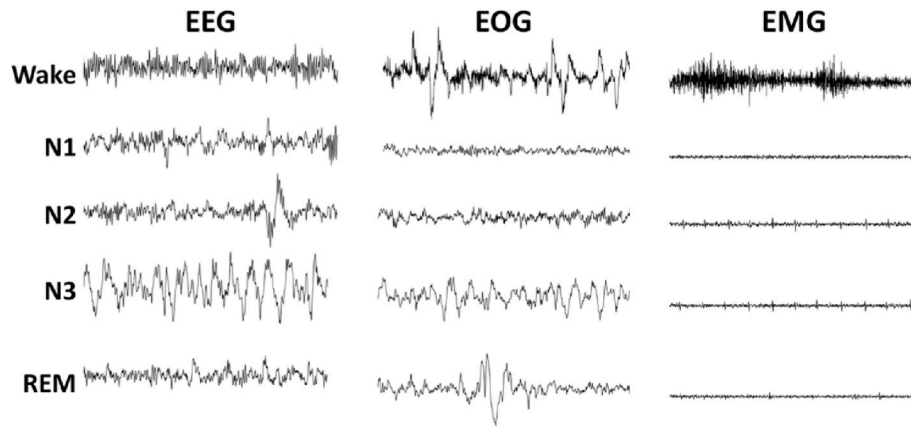


Fig. 1. Typical PSG recordings during the five different sleep stages. Each row shows the EEG, EOG and chin EMG signal during the corresponding sleep stage, respectively.

Table 1

Detailed information of the sleep PSG recording dataset used in this work. “Mov” represented that the epoch contained noises from body movement. “Good” presented good sleep efficiency; “Bad” presented bad sleep efficiency.

Subject group	Total epoch	Wake (%)	N1 (%)	N2 (%)	N3 (%)	REM (%)	Mov (%)
Good	18,026	2.25	3.18	48.98	15.87	19.32	10.40
Bad	19,731	15.40	3.97	36.11	14.70	14.35	15.48
Total	37,757	9.12	3.59	42.26	15.26	16.72	13.05

methods based on traditional machine learning technology have been developed to decrease clinical staff stress and aid in diagnosis [7–9]. Recently, deep learning, a rapidly developing technology, has been successfully applied to image recognition, object detection, semantic segmentation, speech recognition, text analysis, language translation, image generation and other fields [10] and it has also been widely applied to various types of physiological signals [11]. Common model structures in deep learning include convolutional neural networks (CNNs), recurrent neural networks (RNNs), or a combination of CNN and RNN. In the existing literature, many automatic sleep stage methods based on CNN (including one-dimensional CNN and two-dimensional CNN), RNN (including long-short term memory (LSTM) and bidirectional LSTM (BiLSTM)) or the combination of CNN and RNN have been proposed [12–16]. The accuracy of these methods is between 80 and 90%, because of differences in the sleep data between the subjects. These are also caused the agreement between experts was approximately 82% [6].

However, most automatic sleep scoring methods were developed

using sleep data from different subjects instead of personalized sleep data, which might not be suitable for a specific patient condition and obtain high performance, since the changing nature of the EEG signal characteristics are significantly different across patients and strongly depend on the patient’s physical condition, such as age, underlying disease and medication use. The sleep architecture was changed by aging, sleep rhythm and comorbidities of the mental illness [17]. In general, with age progressing in the adult population, the percentage slow wave sleep (SWS) stage and rapid-eye-movement (REM) sleep stage decrease [18]. The prolonged sleep deprivation makes the rebounding of SWS and REM stage [19]. Among patients with depression, they have decreased SWS stage, shorten REM latency, increased length of 1st REM and increased REM density [20]. Therefore, the brain wave during sleep has high heterogeneity between individuals [21,22].

Recent years, the development of precision medicine being actively promoted. The aims of precision medicine is to diagnose an individual’s disease with using the personal data to achieve personal healthcare. The advantage of precision medicine is that it can overcome the inaccuracy of diagnosis caused by individual differences through personal data [23, 24]. One of the first steps towards precision medicine is to understand the disease subtypes. Clinicians can formulate treatment strategies and obtain better treatment results based on the information of each patient, individual differences, and the disease subtypes of the condition [25]. Precision sleep medicine will become an inexorable trend in future.

To develop a personalized sleep scoring classification in precision sleep medicine, considerable time is required to record the sleep data. In addition, the clinical staff also need to spend time monitoring and scoring, which wastes considerable medical costs and resources. Recently, generative adversarial networks (GAN) [26], an advanced

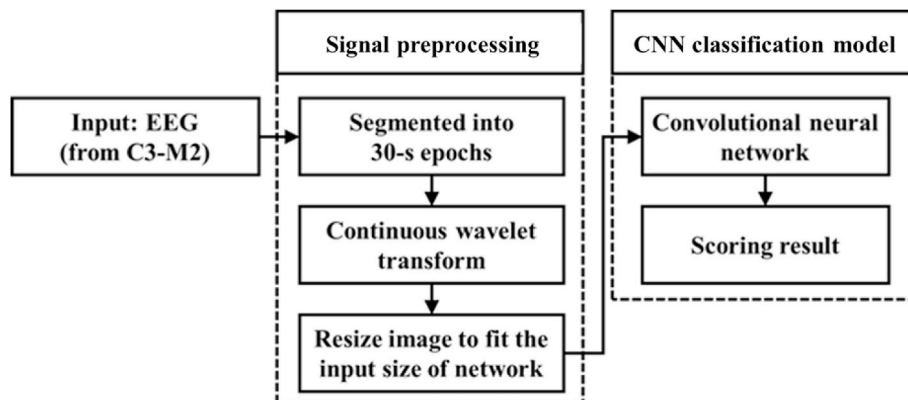


Fig. 2. Block diagram of the proposed automatic sleep scoring method.

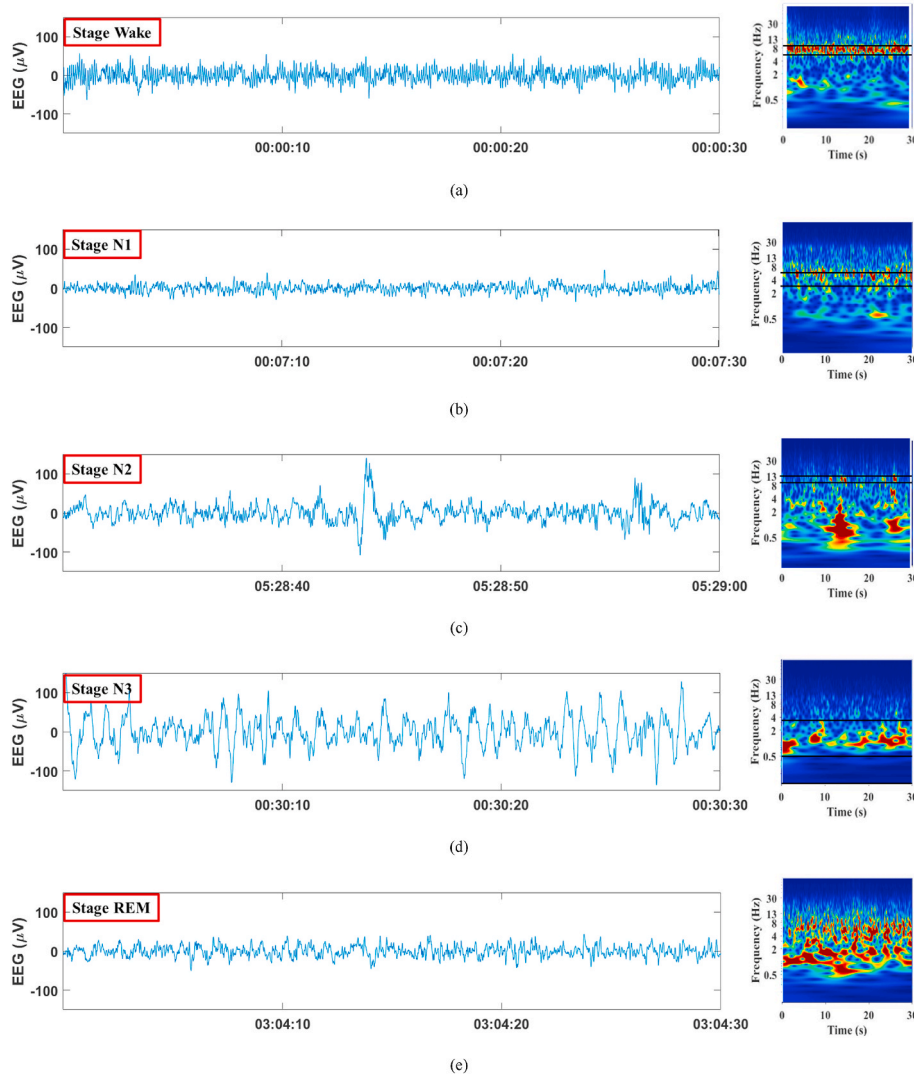


Fig. 3. EEG raw signal at the five different sleep stages ((a) Wake, (b) N1, (c) N2, (d) N3, and (e) REM, respectively) and their corresponding spectrograms.

data augmentation technique, have been successfully applied in the medical imaging domain, such as X-ray, MRI, CT images [27], ECG [28], and EEG [29,30]. Many improved GAN variants, such as Conditional GAN (CGAN), Deep Convolution GAN (DCGAN), Progressive-Growing GAN (PGGAN), Wasserstein GAN (WGAN), and Self-attention GAN (SAGAN) etc., were proposed [31,32]. The adventures of SAGAN are based on attention-driven, long-range dependency modeling for image generation tasks. The details of image can be generated using cues from all feature locations by using SAGAN. In addition, the discriminator of the SAGAN can check that highly detailed features in distant portions of the image are consistent with each other. Therefore, we utilized SAGAN to synthesize more personalized training data to improve the performance of the personalized classification and to reduce the time required for recording, monitoring and scoring.

Our contribution in this study is fourfold. (1) We identify the feasibility of using SAGAN to augment personalized training data to improve the performance of personalized automatic sleep scoring classification. To the best of our knowledge, this is the first application, where the instances generated by an adversarial network have been shown to improve supervised classification tasks outside the synthesized spectrogram of biomedical signals. (2) From the patient side, the proposed method can reduce the total recording times during sleep in the sleep center and decrease the waiting times for measurement to obtain a high accuracy personalized classifier. (3) From the clinical staff side, the

proposed method requires less time for analyzing patients' sleep architectures and evaluating their sleep quality, reducing medical costs and time. (4) We discuss and compare how to set the SAGAN parameter to obtain high-quality personalized training data and the difference between the visualization of the decisions for the trained model with different training data.

2. Materials and methods

2.1. Materials

The dataset contained two groups of subjects, including 19 healthy individuals that were noted as good sleep group (12 males and 7 females; the sleep efficiency $\geq 85\%$) and 23 insomnia patients that were noted as bad sleep group (16 males and 7 females; the sleep efficiency $< 85\%$). The insomnia patients have insomnia more than three days per week lasting for one month. Moreover, the insomnia patients had sleep efficiency less than 85%, sleep onset time more than 15 min and/or wake after sleep onset time more than 30 min according their two-week-actigraphy sleep logs. All measurements were approved by the internal review board of National Cheng Kung University Hospital (NCKUH IRB No. B-ER-100-322). The subjects were publicly recruited by the psychiatric outpatient clinic of the NCKUH. Before the PSG recordings was collected, the subjects were informed that their data was used for sleep-

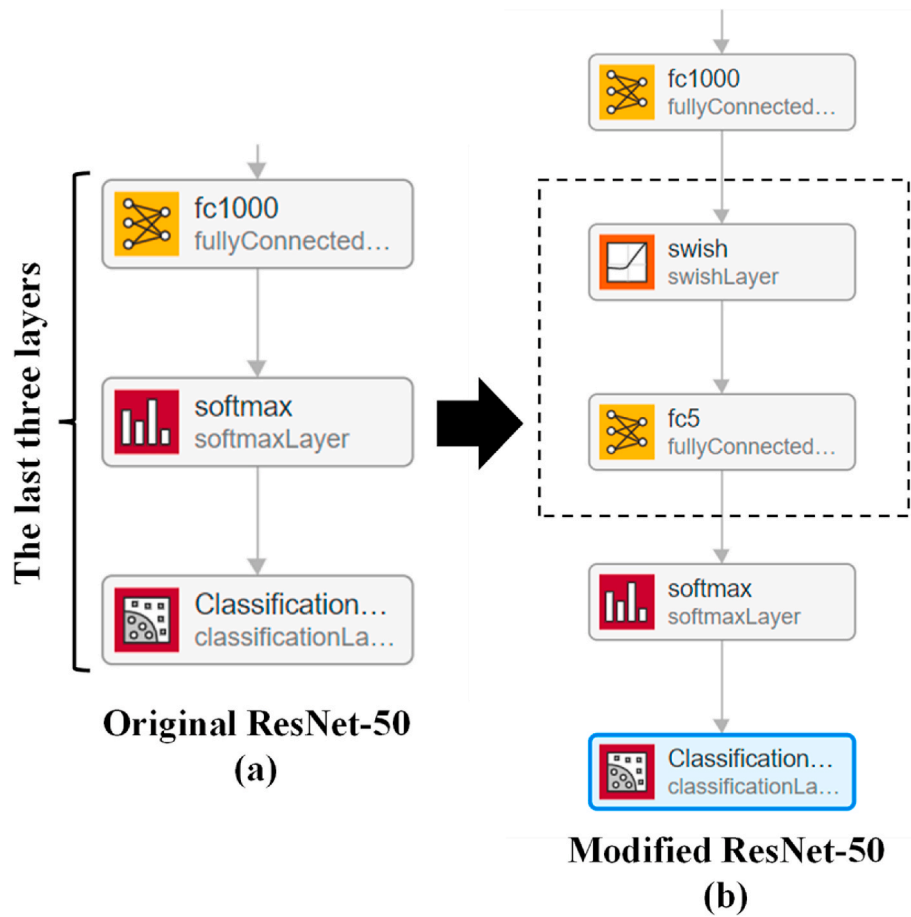


Fig. 4. (A) The last three layers of original ResNet-50; (b) the last three layers of modified ResNet-50, a swish layer and a fully connected layer was added after the last third layer in original ResNet-50.

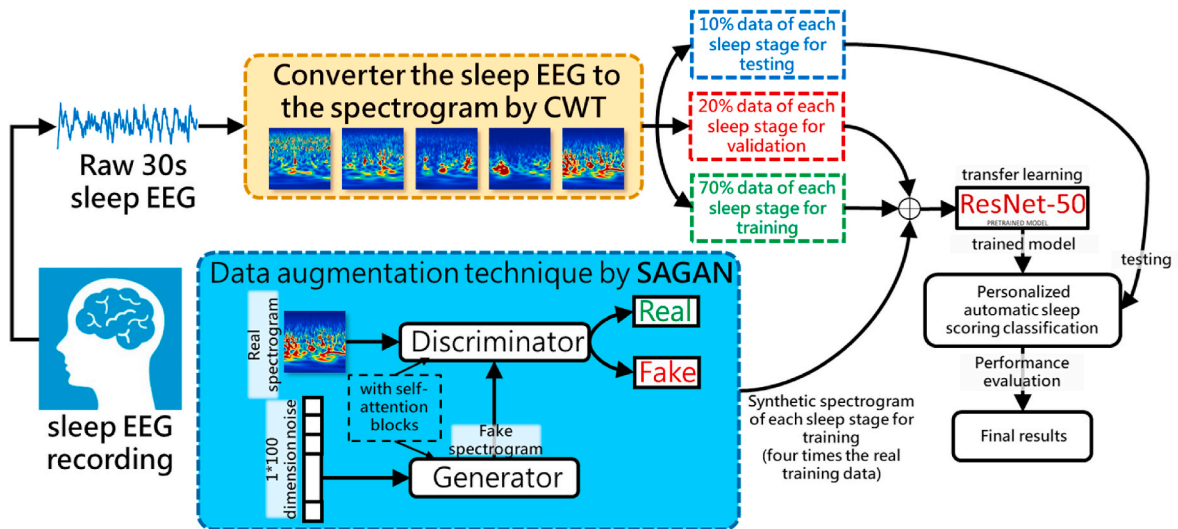


Fig. 5. The architecture of the personalized automatic sleep scoring classifier with using SAGAN.

related research and obtain their consent. The all-night PSGs were recorded in NCKUH. According to AASM rules suggestions and the international 10–20 standard system, the PSG recordings that included C3-M2 EEG and C4-M1 EEG channels, one EOG (ROG-LOG) channel, and a chin EMG channel. The sampling rate of the PSG recordings was setting as 256 Hz. The 42 PSG recordings were manual scored by an expert using the AASM rules. The detailed information about the sleep dataset

was shown in Table 1.

2.2. Automatic sleep scoring method

The block diagram of the automatic sleep scoring method was shown as Fig. 2 that includes two main blocks: 1) signal preprocessing and 2) CNN classification model. The more details of each part were presents as

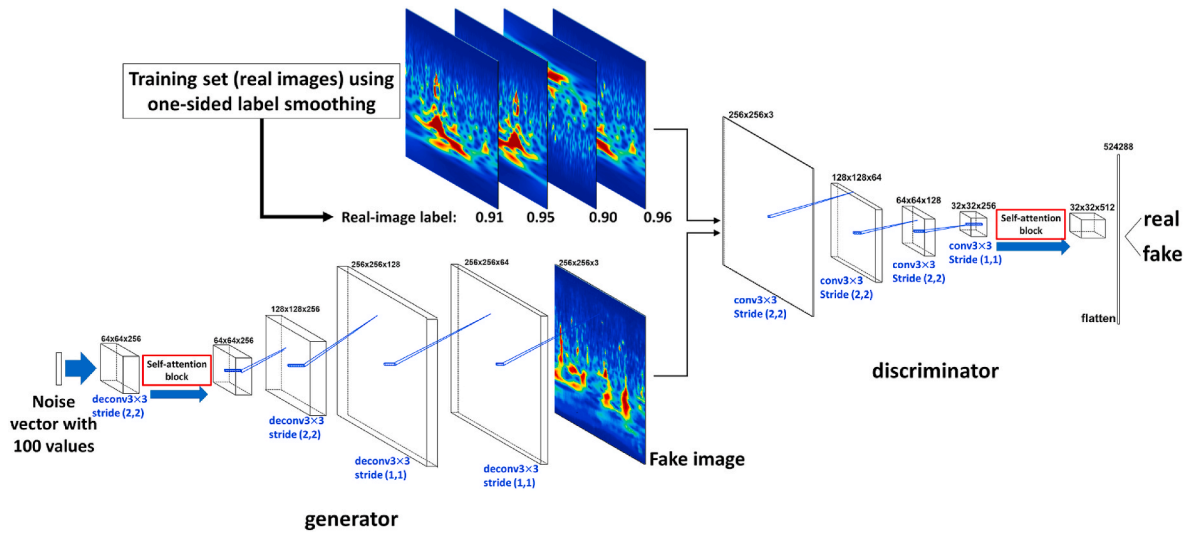
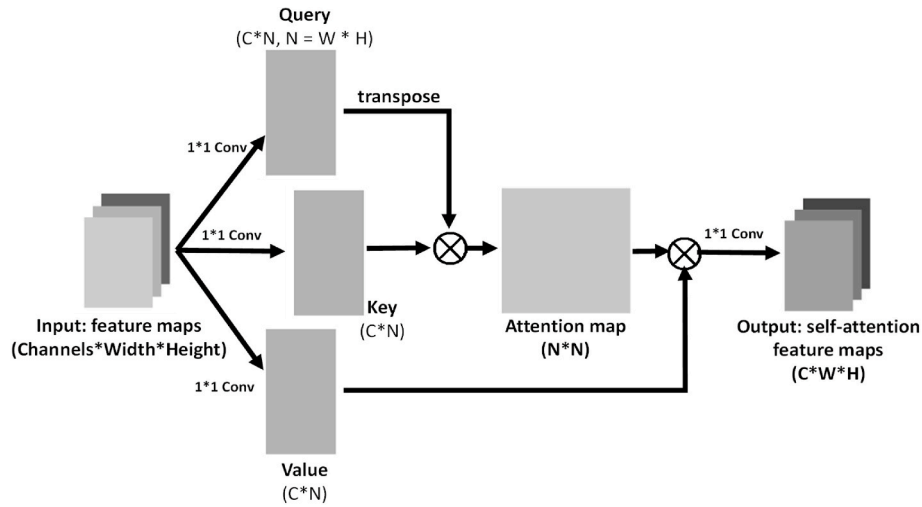


Fig. 6. The architecture of the proposed SAGAN.

Fig. 7. The illustration of the self-attention block for the SAGAN. The $\otimes$ denotes matrix multiplication.

the following figure and descriptions.

1) Signal preprocessing

According to AASM rules suggestions, the overnight EEG signal was segmented into 30-s epochs with non-overlap firstly. Sleep EEG have a variety of characteristics in different frequency bands, such as alpha (8–13 Hz) or delta (0.5–4 Hz) rhythm [33]. If only time-domain or only frequency-domain signal analysis methods are used, the network will learn features with insufficient capabilities. Therefore, we want to convert the signal into a spectrogram so that it has both time domain and frequency domain information at the same time. Common methods of biomedical signal-to-spectrogram methods include short-time Fourier transform (STFT) and continuous wavelet transform (CWT) [34–36]. Compared with the spectrogram based on STFT, the spectrogram based on CWT has more smoothness and continuity. Therefore, we used CWT to convert sleep EEG into a spectrogram images in this study. There are many different mother wavelets can be used in CWT, such as the Morse wavelet or the Morlet wavelet. The Morse wavelet was more suit for EEG since the signal with time-varying amplitude and frequency and we adopted it as the mother wavelet in this study. The following equation (1) shows the CWT formula:

$$W(a, b) = \int_{-\infty}^{\infty} x(t) \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) dt \quad (1)$$

where $x(t)$ is the signal of each 30-s epoch, ψ is a mother wavelet, a is a dilation parameter, and b is a translation parameter. After CWT calculation, the each 30-s epoch was converted to a RGB images for training and testing the classification. The original spectrograms image size was set $256 \times 256 \times 3$, and we resized these images according to the input size of the CNN model.

Fig. 3 shows the raw 30-s EEG and corresponding spectrogram at the Wake (Fig. 3(a)), N1 (Fig. 3(b)), N2 (Fig. 3(c)), N3 (Fig. 3(d)), and REM (Fig. 3(e)) stages, respectively. The X-axis represented time (s) in the spectrogram with the range of 0–30 s. The Y-axis represented frequency (Hz) in the spectrogram and the scale is logarithmic with the range of 0.11–113.11 Hz. The redder the area in the spectrogram means that the power was stronger in the corresponding time and frequency area.

The characteristics of each sleep stage are easily observed in the corresponding spectrogram, such as Fig. 3(a), the main characteristic of the Wake stage is the alpha rhythm. It can be observed that the corresponding spectrogram has larger power in the 8–13 Hz frequency band. The EEG characteristics of N1 and REM are very similar that with mixed frequency (8–13 Hz and 4–8 Hz) also can be observed in the

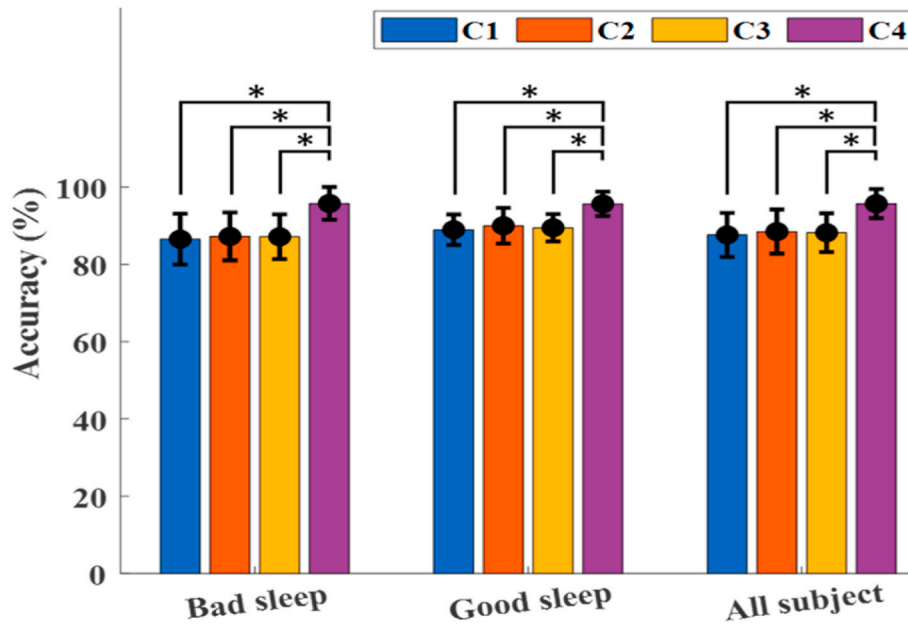


Fig. 8. Average accuracies of the four different classifiers in different group. C1: group classifier without SAGAN; C2: group classifier with SAGAN; C3: personalized classifier without SAGAN; C4: personalized classifier with SAGAN.

spectrogram, as shown in Fig. 3(b) and (e), respectively. The characteristics of the N2 stage are the spindle and K-complex where is the 11–16 Hz, and low frequency respectively, and the frequency band of the 0.5–4 Hz and 11–16 Hz in corresponding spectrogram (see Fig. 3(c)) has larger power. In addition, the main characteristic of the N3 stage is the delta-rhythm wave and the corresponding spectrogram (see Fig. 3(d)) has larger power in the 0.5–4 Hz frequency band. The spectrogram converted by the CWT clearly shows that the power is present in all of the brain rhythms.

2) Classification

Convolutional neural network (CNN), one of the well-known model architectures in deep learning, has been successfully applied to image recognition, object detection, semantic segmentation, speech recognition, image generation and other fields and it has also been widely applied to various types of physiological signals [37]. The main advantage of CNN is that it automatically learn and extract the important features from low to high dimensionality without any feature engineering or human supervision. The applications of various types of physiological signals, such as EEG, ECG, or EMG, had been successfully by using one-dimensional CNN and two-dimensional CNN [38–40]. In this study, the EEG spectrograms are fed to a pre-trained CNN model that named ResNet-50 [41]. We slightly modified the structure of ResNet-50 to make it suitable for EEG spectrogram. The last three layers of original ResNet-50 structure as shown in Fig. 4(a), we add a swish layer [42] (i.e., a nonlinear activation function) and a fully connected layer (the output size is five to fit the category we want to classify) after the last third layer in original ResNet-50, as shown in Fig. 4(b). Swish is a smooth, non-monotonic function that consistently matches or outperforms ReLU on deep networks. In order to make the final features go through a non-linear learning, we add a swish activation function layer instead of just directly connecting the fully connected layer. If the model just adds a fully connected layer directly at the end, in a mathematical sense, it is equivalent to performing matrix multiplication on the features for feature dimension reduction. Compared with traditional transfer learning methods, this method enables the model to relearn features suitable for new tasks through previous learning experience.

2.3. Training data augmentation by SAGANs

Generative adversarial networks (GANs) [43–45] are composed of a generator and a discriminator. The generator, which generates synthesized data given a random noise vector, and the discriminator discriminates actual data and the synthesized data from the generator. To augment the training data for the development of a personalized sleep scoring classification, we applied SAGAN as an advanced data augmentation technique. The advantages of SAGAN are that used the cues from all feature locations to generate the details of image and the highly detailed features in distant portions of the image can be checked by the discriminator of the SAGAN that are consistent with each other or not. Fig. 5 shows the architecture of the personalized automatic sleep scoring classifier with using SAGAN.

For each subject, the randomly selected 70% and 20% of data from each sleep stage was used as training and validation data, respectively. The remaining 10% of data was used as testing data to evaluate the performance of the classifiers trained with different training data. The training data was also used to train SAGAN for each sleep stage. The quantity of synthesized data was four times the real training data (i.e., 280% synthesized data generated from SAGAN), because the ratio of the sleep stage is imbalanced. In order to evaluate and compare the effectiveness of the personalized automatic sleep scoring method with or without using SAGAN, we trained four classifiers for automatic sleep scoring: (1) the group classifier without using SAGAN, (2) the group classifier with using SAGAN, (3) the personalized classifier of each subject without using SAGAN, and (4) the personalized classifier of each subject with using SAGAN. The difference of the four classifiers was their training data. The description is as follows:

- 1. The group classifier without using SAGAN (called C1):** The training data only was 70% random data from each sleep stage of all subjects. It means we use all subjects' training data to train a classifier.
- 2. The group classifier with using SAGAN (called C2):** The training data contained 70% random data from each sleep stage of all subjects and the synthesized data of all subjects. It means we use all subjects' training and synthesized data to train a classifier.
- 3. The personalized classifier without using SAGAN (called C3):** The training data contained 70% random data from each sleep stage

Table 2
Results of the evaluation different classifiers using the group of good and bad sleep efficiency, and all subject. C1: group classifier without SAGAN; C2: group classifier with SAGAN; C3: personalized classifier without SAGAN; C4: personalized classifier with SAGAN.

		average accuracy (%)		F1 score (%)		Se (% , ±SD)		PPV (% , ±SD)							
								Wake		N1		N2		N3	
Bad sleep	C1	86.6 ± 6.6	75.8 ± 9.4	83.8 ± 16.3	22.8 ± 34.7	92.7 ± 5.7	81.3 ± 5.7	88.8 ± 10.0	90.7 ± 13.0	34.4 ± 45.2	87.1 ± 7.7	92.3 ± 15.6	78.9 ± 15.7		
	C2	87.3 ± 6.2	76.0 ± 9.4	90.9 ± 11.8	21.1 ± 37.1	91.8 ± 6.7	87.5 ± 14.0	92.2 ± 8.8	87.3 ± 11.4	30.4 ± 43.6	89.2 ± 6.0	88.3 ± 16.6	76.9 ± 15.0		
	C3	87.2 ± 5.8	77.6 ± 9.2	89.8 ± 16.1	32.8 ± 35.9	90.6 ± 7.7	86.0 ± 15.4	84.7 ± 18.9	97.6 ± 13.3	42.8 ± 40.6	87.3 ± 7.5	87.1 ± 19.5	88.4 ± 9.2		
	C4	95.8 ± 4.24	93.6 ± 7.2	97.1 ± 5.4	80.5 ± 24.4	96.9 ± 3.7	94.8 ± 8.6	95.1 ± 10.7	97.4 ± 4.9	91.4 ± 24.0	95.8 ± 4.7	95.4 ± 6.5	93.8 ± 10.9		
Good sleep	C1	89.0 ± 3.5	74.0 ± 11.1	73.5 ± 40.2	18.7 ± 31.2	92.7 ± 4.9	83.9 ± 15.7	89.6 ± 9.2	74.6 ± 40.6	20.6 ± 34.5	90.6 ± 4.9	88.6 ± 12.2	88.3 ± 10.3		
	C2	90.0 ± 4.6	75.9 ± 9.2	83.2 ± 32.9	12.3 ± 23.0	92.4 ± 4.4	93.4 ± 6.0	93.9 ± 7.7	81.4 ± 32.4	19.3 ± 34.3	92.6 ± 5.4	86.6 ± 12.6	86.7 ± 10.0		
	C3	89.5 ± 3.5	75.9 ± 8.6	87.5 ± 27.5	11.1 ± 17.6	93.8 ± 3.8	84.5 ± 11.7	93.7 ± 5.6	89.1 ± 26.2	26.3 ± 40.9	90.3 ± 4.0	90.9 ± 13.7	86.9 ± 6.8		
	C4	95.7 ± 3.2	92.1 ± 7.6	100.0 ± 0.0	72.7 ± 36.0	97.7 ± 2.4	90.3 ± 11.5	96.3 ± 5.3	97.2 ± 8.5	77.9 ± 36.2	95.8 ± 4.1	97.1 ± 4.5	94.4 ± 6.7		
All subject	C1	87.7 ± 5.7	75.0 ± 10.2	79.2 ± 30.1	18.8 ± 31.2	92.7 ± 4.9	83.9 ± 15.7	89.6 ± 9.2	83.4 ± 30.1	28.1 ± 41.3	88.7 ± 6.8	90.6 ± 6.8	83.2 ± 14.3		
	C2	88.5 ± 5.7	76.0 ± 9.4	87.4 ± 24.1	17.1 ± 31.8	92.1 ± 5.8	90.1 ± 11.5	93.0 ± 8.4	84.63 ± 23.5	25.4 ± 40.1	90.7 ± 6.0	87.5 ± 14.9	81.4 ± 13.8		
	C3	88.2 ± 5.0	76.8 ± 9.0	88.8 ± 22.1	23.0 ± 31.0	92.0 ± 6.4	85.3 ± 13.9	88.8 ± 15.1	88.3 ± 20.2	35.4 ± 41.6	88.7 ± 6.3	88.8 ± 17.2	87.7 ± 8.2		
	C4	95.7 ± 3.8	92.9 ± 7.4	98.4 ± 4.3	77.0 ± 30.4	97.3 ± 3.2	92.8 ± 10.2	95.6 ± 8.7	97.3 ± 6.7	85.3 ± 30.9	95.8 ± 4.4	96.2 ± 5.8	94.1 ± 9.3		

of each subject. It means each subject has himself/herself personal classifier using himself/herself training data to train.

4. **The personalized classifier with using SAGAN (called C4):** The training data contained 70% random data from each sleep stage of each subject and themselves synthesized data generated from the SAGAN. It means each subject has himself/herself personal classifier using himself/herself training and synthesized data to train.

Therefore, in addition to the two group classifiers, each subject had two classifiers, the personalized classifier with and without using SAGAN. The four classifiers were compared to prove that the performance of the personalized classifier is better than the group classifier trained using all subjects' training data, and the synthesized data generated from SAGAN could help to improve the performance of the personalized classifier.

In this study, the EEG spectrograms of the subject were used to train themselves SAGAN to generate the synthesized EEG spectrograms. Then, the actual and synthesized EEG spectrograms of the subject were used as training data to train the personalized classifier for automatic sleep scoring classification. The structure of the proposed SAGAN, which generate data for the group classifier and the personalized classifier, is the same. The difference is the used training data. The training data which contained all subjects' data was used to train the SAGAN, which generated data for group classifier. On the other hand, the training data which contained only one subject's data was used to train the SAGAN, which generated data for personalized classifier.

The architecture of the proposed SAGAN which including the generator and the discriminator structure are shown in Fig. 6. The input layer is a random noise vector that contained 100 values, and the range of the values is -1 to 1 . Then, the input is fed to a dense layer and reshaped to size $64 \times 64 \times 256$. Next, a self-attention block was used to multiply the feature map after the first deconvolution layer and to obtain a more refined and highlighted feature map. The illustration of the self-attention block was shown as Fig. 7. The input is the feature maps and a 1×1 convolution operation was used to generate the query, key and value feature maps, respectively. Next, the attention map was generated by the matrix dot product of query and key features. The attention map describes each pixel's attention score on every other pixel. The self-attention feature maps were obtained by the matrix dot product of value feature maps and attention map. The self-attention feature maps describe each pixel's total attention score throughout all pixels. We then reshape the self-attention feature maps into the shape of $C \times W \times H$. For a more detailed description of the self-attention block, please refer to [32].

After the self-attention block, the self-attention feature maps were fed to three deconvolution layers to generate a synthesized image. Batch normalization and ReLU activation function are performed on each layer of the generator, except for the output layer. The output layer uses a tanh activation function. The discriminator is a CNN structure which also had a self-attention block. The input image size of the discriminator is $256 \times 256 \times 3$, and the output is a decision: actual or fake image. Four convolution layers with a kernel size of 5×5 and the last convolution layers was multiplied by a self-attention block. In addition, a dense layer was contained. Each convolution layer uses a stride to reduce the spatial dimensionality and uses Leaky ReLU activation function with a slope of 0.2 . The output layer uses the sigmoid activation function for the probability of the image.

The model hyperparameters of the generator and discriminator was manually tuned from different architectures, including the kernel size was set 3×3 , 5×5 , and 7×7 , the stride was set 1×1 and 2×2 , and the number of convolution and deconvolution layers was set 3, 4, and 5, respectively. The optimizer chose adaptive moment estimation (Adam) to train the SAGAN, the batch size was set 32 to avoid the images in a training batch were too similar that made mode collapse which is a frequently problem when training GAN, and the training interaction was set 1000 to prevent over learning. The main issue of generating

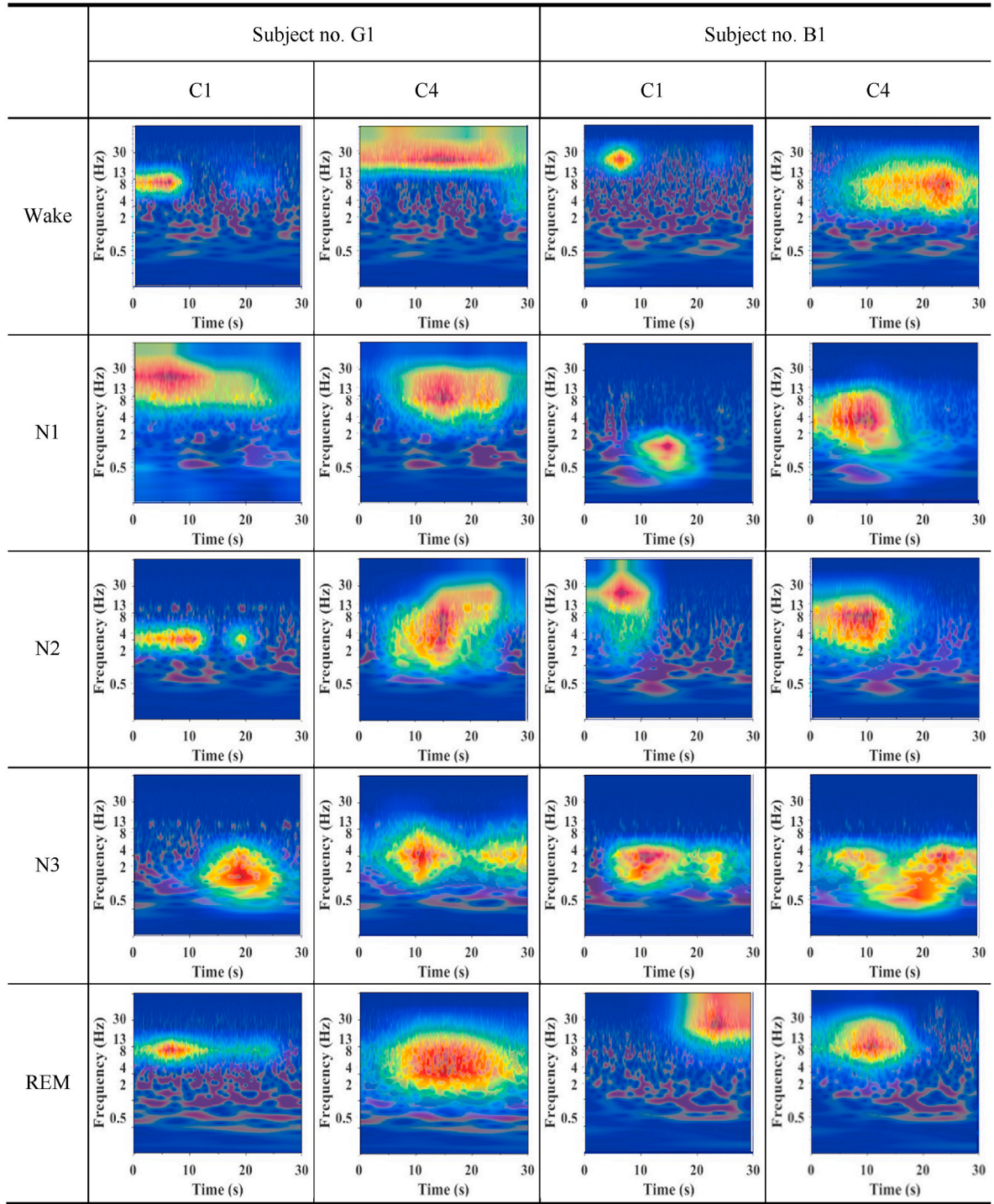


Fig. 9. Decision as a heat map of C1 and C4 in each sleep stage.

personalized data is the difference in the same sleep stage is usually not obvious, and thus it is easy to lead to mode collapse. In order to prevent the issue, we used a special one-sided label smoothing technique [46] to prevent mode collapse, add the diversity of the generated data, and retain the personalized data style. One-sided label smoothing is an approach to provide smoothed labels for the discriminator that avoid mode collapse. The common way of one-sided label smoothing is used a constant value such as 0.9, instead of 1. However, the training data of the personalized SAGAN is usually similar, it will often lead to mode collapse even using one-sided label smoothing. Therefore, we used the special one-sided label smoothing that set the labels of the real data is

the randomly values from 0.9 to 1 to solve the above problem. The main idea of the special one-sided label smoothing is to make the synthesized EEG spectrograms close to the real EEG spectrograms with the personalized data style, but not too close. Therefore, the special one-sided label smoothing can synchronously prevent mode collapse and retain the personalized data style.

2.4. Performance evaluation

The four classifiers were trained using different training data. Confusion matrices were used to evaluate the sleep scoring classification

Table 3

Cooperation of our proposed method with the related study of the automatic sleep scoring.

Reference	Dataset & Subjects	Classification method	Performance (%)
[48]	SHHS/5793 subjects	raw signal + CNN	Acc: 87.00 F1: 78.00
[49]	Sleep-EDF/42 healthy subjects	raw signal + CNN + LSTM	Acc: 84.26 F1: 79.66
[50]	sleep-EDF/12 healthy subjects	55 features + ensemble RNN	Acc: 86.74 F1: unreported
[51]	UCD/25 subjects	spectrograms + orthogonal CNN	Acc: 88.40 F1: unreported
[52]	Sleep-EDF/22 healthy subjects	raw signal + CNN	Acc: 83.36 F1: 79.00
[53]	Sleep-EDF/20 healthy subjects	raw signal + CNN + LSTM	Acc: 82.02 F1: 76.90
[54]	MASS/62 healthy subjects	time-frequency series + MLP + LSTM	Acc: 85.92 F1: 80.50
[55]	Sleep-EDF/19 healthy subjects	raw signal + CNN	Acc: 84.00 F1: unreported
proposed method	42 subjects (23 healthy & 19 patient)	spectrograms + CNN (C1)	Acc: 87.70 F1: 75.00
		spectrograms + GAN + CNN (C4)	Acc: 95.70 F1: 92.90

performance. We calculated the accuracy (Acc), sensitivity (Se), positive predictive value (PPV) and the F1 score. for each personalized classification, respectively. All measures are presented in the following equations:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Se = \frac{TP}{TP + FN} \quad (3)$$

$$PPV = \frac{TP}{TP + FP} \quad (4)$$

$$F_1 = \frac{2 \times (Se \times PPV)}{Se + PPV} \quad (5)$$

where TP , TN , FP , and FN were represent true positive, true negative, false positive, and false negative, respectively. Se represents that the proportion of actual positives cases that are correctly identified. PPV represents that the percentage of correct predictions. F1 Score is the weighted average of Se and PPV , that usually more useful than accuracy, especially when the classes are not evenly distributed.

3. Results

Fig. 8 shows the average accuracies of the four different classifiers. The average accuracies and standard deviations (SD) of C1, C2, C3, and C4 in all subjects were 87.66% (± 5.68), 88.52 (± 5.71), 88.23% (± 5.01), and 95.74% (± 3.78), respectively. In the group classifier with and

without using SAGAN, the average accuracies of the group classifier with using SAGAN was better than that without using SAGAN. After using SAGAN, the average accuracy was separately increased by 0.7%, 1.1%, and 0.9% in good sleep, bad sleep, and all subjects, respectively. The result proved that data augmentation using SAGAN can improve the performance of the classification. In the other hand, the group classifier with data augmentation using SAGAN is unable to improve the performance a lot, because the characteristics of different sleep stage may be similar among the subjects.

In the case of not using SAGAN (i.e., C1 and C3), the average accuracy of the personalized classifier was better than that of the group classifier but did not significantly improve. The difference in the average accuracy between the group classifier and the personalized classifier was 0.62%, 0.52%, and 0.57% in good sleep, bad sleep, and all subjects, respectively. We assumed that the quantity of personalized training data was not sufficient when training the personalized classifier. Thus, the performance was not significantly improved. In addition, the SD was decreased using the personalized classifier even with or without using SAGAN. This proved that the group classifier cannot scale well across different individuals.

The performance of the personalized classifier with using SAGAN (i.e., C4) had the highest average accuracy. The average accuracy of the personalized classifier with using SAGAN reached 95.8%, 95.67%, and 95.74% in good sleep, bad sleep, and all subjects, respectively. In addition, a significant difference can be observed with one-way analysis of variance between the personalized classifier with using SAGAN and the other classifiers. The results indicated that data augmentation can improve the performance of the personalized classifier, and SAGAN applies successfully in data augmentation.

The results, F1 score of each classifier, and Se and PPV of each sleep stage in different groups are shown in Table 2. The Se and PPV of N1 are the lowest in automatic sleep scoring, because the ratio of the N1 stage in the sleep cycle is the lowest (see Table 1). We observe that the personalized classifier with using SAGAN can significantly improve the Se of N1 and the SD of each sleep stage. The results indicated that the group classifier with a large SD might not be suitable for direct application in specific individuals because of individual differences. Although the personalized classifier can improve the accuracy of automatic sleep scoring method, it needs sufficient amount of personalized training data to train the personalized classifier. Achieving more training data requires more PSG recording, which increases the medical costs and time for recording, monitoring and scoring. The ability of the generative models to add significant amounts of training examples carries higher promise.

4. Discussions

First, we compare the differences between the C1 and C2 groups to understand the impact on the accuracy of the classifier after data augmentation using the SAGAN that trained with the group data, i.e., non-personalized data. According to Table 2, each evaluation index of

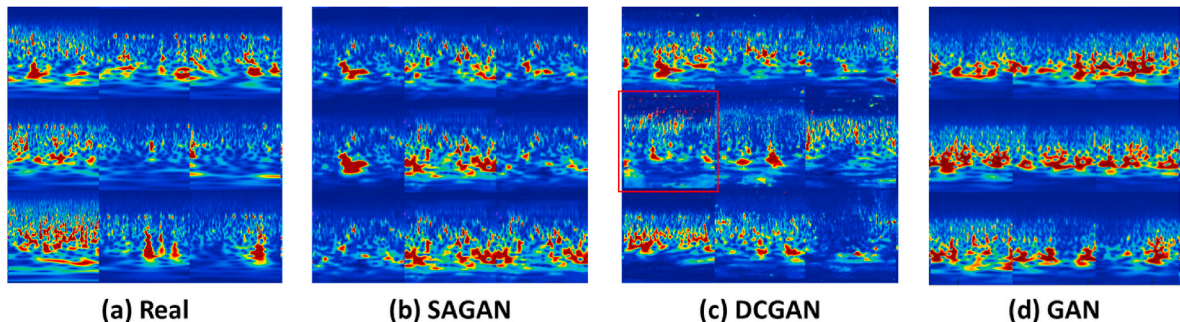


Fig. 10. The real EEG spectrograms (a) and the synthesized EEG spectrograms that generated by SAGAN (b), DCGAN (c), and GAN (d), respectively.

C2 group, whether it is bad sleep, good sleep, or the all-subject groups, has a slight improvement compared with the C1 group. The accuracies of bad sleep group, good sleep group, and the all-subject groups increase more than 0.7%, 1%, and 0.8%, respectively. The reason is that the SAGAN trained by using the group data, its function is simply to increase the overall training data, and didn't take into account the problem of individual differences. In other words, the data generated by the SAGAN is mixed with the characteristics of multiple individuals, so the accuracy is not significantly improved.

Compared with other groups, the accuracy of C4 group is significantly improved. This is because the SAGAN trained by personalized data not only increases the overall training data, but also generates data with individual characteristics. Therefore, the accuracy can be improved significantly. If the trained model has too high accuracy, it may be considered whether it is overfitting. In general, the ideal model should be general, that is, the model has good accuracy across different groups of test data. If the model can only have extremely high accuracy in one specific group, but other groups are relatively low, we will consider the model to be overfitting. However, in the definition of precision medicine, each medical strategy or behavior is personalized. Each model that trained by personalized data will only be used by himself, so there is no doubt about whether the model is overfitting. Moreover, the results of C4 group imply our proposed SAGAN can increase the personalized training data successfully and get the personal model with high accuracy. This not only improves the accuracy of medical diagnosis, but also greatly reduces the additional time spent by medical personnel in collecting and scoring patient data, and the medical burden. On the other hand, patients can also save time and money on additional data collection.

To further understand the difference in the decision between C1 and C4, we use a visualization tool, called Grad CAM [47], to present the decision as a heat map of C1 and C4. Fig. 9 shows outputs from C1 and C4 in subject G1 and B1 with each sleep stage, respectively. In the heat map, the red regions correspond to high scores for the sleep stage; the blue regions correspond to low scores for the sleep stage.

Before using SAGAN (i.e., C1), the red region of the heat map was smaller. This means that C1 classified a spectrogram into a specific sleep stage based on only a small area in the spectrogram. It shows that C1 uses a small amount of information to determine the classification result. However, in the condition of the clinical staff manual scoring, the classification of sleep stages is based on the 30-s EEG signals instead of a small interval. Therefore, the decision of the C1 classification may be less reliable.

In contrast to C1, after using SAGAN (i.e., C4), the red region of the heat map was larger in each spectrogram of five sleep stages. This means that C4 classified a spectrogram into a specific sleep stage based on most of the area in the spectrogram. It shows that C4 uses a large amount of information to determine the classification result. In addition, the location of the red region contained the characteristic that was referred to in manual scoring. Therefore, the decision of the C4 classification is closer to the condition of the clinical staff manual scoring and more reliable. The results also showed that the decision ability of the personalized classifier could be improved by personalized training data generated from GAN. Summary, because of individual difference, C1 classifier could not clearly identify the important of region. The personalized classifier could be that the important of region are more clearly identified.

We compared our proposed method with the related study of the automatic sleep scoring, as showing Table 3. Some studies used the same dataset (sleep-EDF) but the number of subjects is different. Those studies were also used single EEG and deep learning technique, and their accuracies and F1 scores were in the range of, 85%–90% and 76%–80.5%, respectively. Mikkelsen et al. [55] trained a sleep scoring model using all subjects, and then fine-tuned the model using an all-night PSG data of the subject to build a personal model. The difference between [55] and our proposed method is that the presence or absence of SAGAN. The

performance of the group classifier (C1) in our proposed method also had a similar level (Acc: 87.70% and F1: 75%). The performance of the personalized classifier with SAGAN (C4) was higher (Acc: 95.70% and F1: 92.90%) than the group classifier (C1) and the related studies.

In order to compare the synthesized EEG spectrograms that generated by different GANs, we also applied deep convolution GAN (DCGAN) [45] and original GAN [26] to generate the synthesized EEG spectrograms. Fig. 10 (a)–(d) shows the real EEG spectrograms and the synthesized EEG spectrograms that generated by SAGAN, DCGAN, and GAN, respectively. The spectrograms that generated by GAN obviously lack diversity. Moreover, the spectrograms that generated by DCGAN have diversity but sometimes it will generate obvious fake spectrograms (such as in Fig. 10 (c) that marked by red box). The spectrograms that generated by SAGAN were close to the real spectrograms and have diversity at the same time. The synthesized EEG spectrograms generated by DCGAN and GAN were also used to train and evaluated the C4 classifier by same experimental process, respectively. The average accuracies of the C4 classifier that trained with the synthesized spectrograms generated by DCGAN and GAN were 87.6% and 90.2%, respectively (95.7% by SAGAN). The results proved that SAGAN is very powerful and successful data augmentation technique for developing the personalized automatic sleep scoring classification.

5. Conclusion

In this study, SAGAN was used to synthesize more personalized training data to improve the performance of the personalized classification. We evaluated the performance on the personalized classifier for sleep stage scoring classification and showed that it achieved high performance compared with the group classifier. The average accuracy and standard deviation of the proposed method are 95.74% and 3.78%, respectively. Compared to the classifier trained with all subjects' training data, the average accuracy increased by 8.08%. To the best of our knowledge, this is the first application, where the instances generated by an adversarial network have been shown to improve supervised classification tasks outside the synthesized spectrogram of biomedical signals. In addition, the GAN-based data augmentation method can reduce the time for recording, monitoring and scoring which requires approximately 24–30 h (3 days $\times$ 8–10 h per day) for each subject. In contrast, the total time for training five SAGANs to generate the personalized training data of the five sleep-stage is only approximately 2.5 h. In the future, more personalized data, such as different ages, brain diseases, and sleep disorders, need to be evaluated for the robustness of the SAGAN-based data augmentation method.

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References

- [1] H. Phan, F. Andreotti, N. Cooray, O.Y. Chén, M. De Vos, Joint classification and prediction CNN framework for automatic sleep stage classification, *IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng.* 66 (2018) 1285–1296.
- [2] C.-E. Kuo, G.-T. Chen, P.-Y. Liao, An EEG spectrogram-based automatic sleep stage scoring method via data augmentation, ensemble convolution neural network, and expert knowledge, *Biomed. Signal Process Control* 70 (2021), 102981.
- [3] A. Rechtschaffen, A Manual for Standardized Terminology, Techniques and Scoring System for Sleep Stages in Human Subjects, Brain information service, 1968.
- [4] R.B. Berry, R. Brooks, C. Gamaldo, S.M. Harding, R.M. Lloyd, S.F. Quan, et al., *AASM Scoring Manual Updates for 2017 (Version 2.4)*, American Academy of Sleep Medicine, 2017.
- [5] S.-F. Liang, Y.-H. Shih, P.-Y. Chen, C.-E. Kuo, Development of a human-computer collaborative sleep scoring system for polysomnography recordings, *PLoS One* 14 (2019), e0218948.

- [6] R.S. Rosenberg, S. Van Hout, The American Academy of Sleep Medicine inter-scoring reliability program: sleep stage scoring, *J. Clin. Sleep Med.* 9 (2013) 81–87.
- [7] L. Fiorillo, A. Puiatti, M. Papandrea, P.L. Ratti, P. Favaro, C. Roth, et al., Automated sleep scoring: a review of the latest approaches, *Sleep Med. Rev.* 48 (2019), 101204.
- [8] A. Krakovská, K. Mezeiová, Automatic sleep scoring: a search for an optimal combination of measures, *Artif. Intell. Med.* 53 (2011) 25–33.
- [9] K. Šušmáková, A. Krakovská, Discrimination ability of individual measures used in sleep stages classification, *Artif. Intell. Med.* 44 (2008) 261–277.
- [10] S. Pouyanfar, S. Sadiq, Y. Yan, H. Tian, Y. Tao, M.P. Reyes, et al., A survey on deep learning: algorithms, techniques, and applications, *ACM Comput. Surv.* 51 (2018) 1–36.
- [11] O. Faust, Y. Hagiwara, T.J. Hong, O.S. Lih, U.R. Acharya, Deep learning for healthcare applications based on physiological signals: a review, *Comput. Methods Progr. Biomed.* 161 (2018) 1–13.
- [12] A. Patanaik, J.L. Ong, J.J. Gooley, S. Ancoli-Israel, M.W.L. Chee, An end-to-end framework for real-time automatic sleep stage classification, *Sleep* 41 (2018), zsy041.
- [13] H. Sun, W. Ganglberger, E. Panneerselvam, M.J. Leone, S.A. Quadri, B. Goparaju, et al., Sleep staging from electrocardiography and respiration with deep learning, *Sleep* (2019), zsz306.
- [14] L. Zhang, D. Fabbri, R. Upender, D. Kent, Automated sleep stage scoring of the Sleep Heart Health Study using deep neural networks, *Sleep* 42 (2019) zsz159.
- [15] O. Faust, H. Razaghi, R. Barika, E.J. Ciaccio, U.R. Acharya, A review of automated sleep stage scoring based on physiological signals for the new millennia, *Comput. Methods Progr. Biomed.* 176 (2019) 81–91.
- [16] Phan H, Mikkelsen K, Chén OY, Koch P, Mertins A, De Vos M. SleepTransformer: Automatic Sleep Staging with Interpretability and Uncertainty Quantification. *arXiv preprint arXiv:210511043*. 2021.
- [17] J.K. Paulose, C. Wang, B.F. O'Hara, V.M. Cassone, The effects of aging on sleep parameters in a healthy, melatonin-competent mouse model, *Nat. Sci. Sleep* 11 (2019) 113.
- [18] J. Li, M.V. Vitiello, N.S. Gooneratne, Sleep in normal aging, *Sleep med. clinic.* 13 (2018) 1–11.
- [19] L. De Gennaro, M. Ferrara, M. Bertini, The relationship between frequency of rapid eye movements in REM sleep and SWS rebound, *J. Sleep Res.* 9 (2000) 155–159.
- [20] N.F. Zaki, D.W. Spence, A.S. BaHammam, S.R. Pandi-Perumal, D.P. Cardinali, G. M. Brown, Chronobiological theories of mood disorder, *Eur. Arch. Psychiatr. Clin. Neurosci.* 268 (2018) 107–118.
- [21] B. Moezzi, L.M. Pratti, B. Hordacre, L. Graetz, C. Berryman, L.M. Lavrencic, et al., Characterization of young and old adult brains: an EEG functional connectivity analysis, *Neuroscience* 422 (2019) 230–239.
- [22] B.A. Mander, J.R. Winer, M.P. Walker, Sleep and human aging, *Neuron* 94 (2017) 19–36.
- [23] D.C. Lim, D.R. Mazzotti, K. Sutherland, J.W. Mindel, J. Kim, P.A. Cistulli, et al., Reinventing polysomnography in the age of precision medicine, *Sleep Med. Rev.* 52 (2020), 101313.
- [24] M. Bosi, A. De Vito, B. Kotecha, L. Viglietta, A. Braghiroli, J. Steier, et al., Phenotyping the pathophysiology of obstructive sleep apnea using polygraphy/polysomnography: a review of the literature, *Sleep Breath.* 22 (2018) 579–592.
- [25] J.C. Carberry, J. Amatoory, D.J. Eckert, Personalized management approach for OSA, *Chest* 153 (2018) 744–755.
- [26] I.J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, et al., Generative Adversarial Networks. *arXiv Preprint arXiv:14062661*, 2014.
- [27] X. Yi, E. Walia, P. Babyn, Generative adversarial network in medical imaging: a review, *Med. Image Anal.* (2019), 101552.
- [28] T. Golany, K. Radinsky, PGANS: Personalized Generative Adversarial Networks for ECG Synthesis to Improve Patient-specific Deep ECG Classification, AAAI, Honolulu, Hawaii, USA, 2019, pp. 557–564.
- [29] S.M. Abdelfattah, G.M. Abdelrahman, M. Wang, Augmenting the size of EEG datasets using generative adversarial networks, in: 2018 International Joint Conference on Neural Networks (IJCNN), IEEE, Rio de Janeiro, Brazil, 2018, pp. 1–6.
- [30] K.G. Hartmann, R.T. Schirmer, T. Ball, EEG-GAN: Generative Adversarial Networks for Electroencephalographic (EEG) Brain Signals. *arXiv Preprint arXiv:180601875*, 2018.
- [31] A. Jabbar, X. Li, B. Omar, A Survey on Generative Adversarial Networks: Variants, Applications, and Training. *arXiv Preprint arXiv:200605132*, 2020.
- [32] H. Zhang, I. Goodfellow, D. Metaxas, A. Odena, Self-attention generative adversarial networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 7354–7363.
- [33] S.-F. Liang, C.-E. Kuo, Y.-H. Hu, Y.-H. Pan, Y.-H. Wang, Automatic stage scoring of single-channel sleep EEG by using multiscale entropy and autoregressive models, *IEEE Trans. Instrum. Meas.* 61 (2012) 1649–1657.
- [34] O.H. Colak, Preprocessing effects in time-frequency distributions and spectral analysis of heart rate variability, *Digit. Signal Process.* 19 (2009) 731–739.
- [35] M. Wacker, H. Witte, Time-frequency techniques in biomedical signal analysis, *Methods Inf. Med.* 52 (2013) 279–296.
- [36] R. Boostani, F. Karimzadeh, M. Nami, A comparative review on sleep stage classification methods in patients and healthy individuals, *Comput. Methods Progr. Biomed.* 140 (2017) 77–91.
- [37] A. Krizhevsky, I. Sutskever, G.E. Hinton, ImageNet classification with deep convolutional neural networks, in: Proceedings of the 25th International Conference on Neural Information Processing Systems - Volume 1, Curran Associates Inc., Lake Tahoe, Nevada, 2012, pp. 1097–1105.
- [38] H. Phan, F. Andreotti, N. Cooray, O.Y. Chén, M. De Vos, SeqSleepNet: end-to-end hierarchical recurrent neural network for sequence-to-sequence automatic sleep staging, *IEEE Trans. Neural Syst. Rehabil. Eng.* 27 (2019) 400–410.
- [39] A. Phinyomark, E. Scheme, EMG pattern recognition in the era of big data and deep learning, *Big Data and Cognit. Comp.* 2 (2018) 21.
- [40] S.M. Mathews, C. Kambhampettu, K.E. Barner, A novel application of deep learning for single-lead ECG classification, *Comput. Biol. Med.* 99 (2018) 53–62.
- [41] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [42] P. Ramachandran, B. Zoph, Q.V. Le, Searching for Activation Functions. *arXiv Preprint arXiv:171005941*, 2017.
- [43] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, et al., Generative adversarial nets. *Advances in neural information processing systems*. Palais des Congrès de Montréal, Montréal CANADA: Neural Information Processing Systems Foundation, Inc., 2014, pp. 2672–2680.
- [44] A. Creswell, T. White, V. Dumoulin, K. Arulkumaran, B. Sengupta, A.A. Bharath, Generative adversarial networks: an overview, *IEEE Signal Process. Mag.* 35 (2018) 53–65.
- [45] A. Radford, L. Metz, S. Chintala, Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks. *arXiv Preprint arXiv:151106434*, 2015.
- [46] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, X. Chen, Improved Techniques for Training Gans. *arXiv Preprint arXiv:160603498*, 2016.
- [47] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-CAM: visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE International Conference on Computer Vision, IEEE, Venice, Italy, 2017, pp. 618–626.
- [48] A. Sors, S. Bonnet, S. Mirek, L. Vercueil, J.-F. Payen, A convolutional neural network for sleep stage scoring from raw single-channel EEG, *Biomed. Signal Process Control* 42 (2018) 107–114.
- [49] S. Mousavi, F. Afghah, U.R. Acharya, SleepEEGNet: automated sleep stage scoring with sequence to sequence deep learning approach, *PLoS One* 14 (2019).
- [50] N. Michielli, U.R. Acharya, F. Molinari, Cascaded LSTM recurrent neural network for automated sleep stage classification using single-channel EEG signals, *Comput. Biol. Med.* 106 (2019) 71–81.
- [51] J. Zhang, R. Yao, W. Ge, J. Gao, Orthogonal convolutional neural networks for automatic sleep stage classification based on single-channel EEG, *Comput. Methods Progr. Biomed.* 183 (2020), 105089.
- [52] M. Perslev, M. Jensen, S. Darkner, P.J. Jennum, C. Igel, U-Time: A Fully Convolutional Network for Time Series Segmentation Applied to Sleep Staging. *Advances in Neural Information Processing Systems*, 2019, pp. 4417–4428.
- [53] A. Supratak, H. Dong, C. Wu, Y. Guo, DeepSleepNet: a model for automatic sleep stage scoring based on raw single-channel EEG, *IEEE Trans. Neural Syst. Rehabil. Eng.* 25 (2017) 1998–2008.
- [54] H. Dong, A. Supratak, W. Pan, C. Wu, P.M. Matthews, Y. Guo, Mixed neural network approach for temporal sleep stage classification, *IEEE Trans. Neural Syst. Rehabil. Eng.* 26 (2017) 324–333.
- [55] K. Mikkelsen, M. De Vos, Personalizing Deep Learning Models for Automatic Sleep Staging. *arXiv Preprint arXiv:180102645*, 2018.